# The fibers of a half adjoint equivalence are contractible

The Univalent Foundations Program

hott-book:equivalences.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

*Editorial note.* The counted statement is the author's 'thm:contr-hae' (source lines 265-267); the proof environment starts on the next line, so the pairing is by adjacency.

*Editorial note.* The retained context is the author's characterisation of paths in a fiber, invoked at the first step of the retained proof; retaining it keeps that label defined inside the package.

*Editorial note.* The retained counter values make the counted result print with its original book number, and the chapter and section counters are set so that the retained context (where present) prints with its own original number as well.

*Editorial note.* This source directory intentionally contains only the author file, so the retained book macro library is bundled as a per-task copy inside this task directory and referenced from preamble_spans; the copy is verbatim from the same pinned book snapshot and no macro is added, changed or invented.

*Editorial scope.* One original result of the HoTT Book, counted once: fibers of a half adjoint equivalence are contractible (author label thm:contr-hae).

*Allowed standard prerequisites (editorial).*

- Half adjoint equivalence. ishae(f) consists of g, eta : g o f ~ id, epsilon : f o g ~ id and the coherence tau : prd x, f(eta x) = epsilon(f x).. Its four components are the data used throughout the proof.
- Paths in a fiber. (x,p) = (x',p') in the fiber of f at y is equivalent to Sigma gamma : x = x', ap f gamma 'ct' p' = p.. It is the retained characterisation applied at source line 272.
- Naturality of epsilon. The homotopy epsilon satisfies its naturality identity on paths.. It is the last equality of the displayed chain at source line 280.

**Lemma 4.2.5.** *For any $f : A \to B$, $y : B$, and $(x, p), (x', p') : \mathsf{fib}_f(y)$, we have*

$$\big((x, p) = (x', p')\big) \simeq \Big(\sum_{\gamma : x = x'} f(\gamma) \cdot p' = p\Big)$$

**Theorem 4.2.6.** *If $f : A \to B$ is a half adjoint equivalence, then for any $y : B$ the fiber $\mathsf{fib}_f(y)$ is contractible.*

*Proof.* Let $(g, \eta, \epsilon, \tau) : \mathsf{ishae}(f)$, and fix $y : B$. As our center of contraction for $\mathsf{fib}_f(y)$ we choose $(gy, \epsilon y)$. Now take any $(x, p) : \mathsf{fib}_f(y)$; we want to construct a path from $(gy, \epsilon y)$ to $(x, p)$. By Lemma 4.2.5, it suffices to give a path $\gamma : gy = x$ such that $f(\gamma) \cdot p = \epsilon y$. We put $\gamma :\equiv g(p)^{-1} \cdot \eta x$. Then we have

$$\begin{aligned} f(\gamma) \cdot p &= fg(p)^{-1} \cdot f(\eta x) \cdot p \\ &= fg(p)^{-1} \cdot \epsilon(fx) \cdot p \\ &= \epsilon y \end{aligned}$$

where the second equality follows by $\tau x$ and the third equality is naturality of $\epsilon$. □